

Business Requirements Document

AI Front-Desk Agent for Solo & Small Real Estate Brokers — MVP (Phase 1, Single Test Client)

Product	Real Estate Lead Intake & Follow-Up Agent
Document Version	1.0
Status	Draft — MVP scope, single test client
Prepared For	Broker / Brokerage Owner
Delivery Vehicle	Google Apps Script + Google Forms/Sheets + Twilio + Claude API

1. Executive Summary

Solo and small real estate brokers lose leads to slow response times: a lead who fills out a web form and doesn't hear back within minutes frequently moves on to a competitor or a portal that responds instantly. This project delivers an AI-assisted front-desk system that captures leads from a web form, contacts them by SMS within 60 seconds, follows up automatically on a reminder cadence, and drafts AI-assisted replies to inbound questions for the broker to review and approve — without ever auto-sending unreviewed AI content to a lead.

2. Business Objectives

- Reduce lead first-response time from hours to under 60 seconds.
- Prevent leads from going cold by running an automatic, time-boxed reminder sequence.
- Give the broker a single dashboard view of lead status, last contact, and next scheduled action.
- Keep the broker in control of anything AI-drafted before it reaches a lead (Phase 1 trust model).
- Prove the model end-to-end for one test client before generalizing to a multi-tenant offering.

3. Scope

3.1 In Scope (Phase 1 MVP)

- Web-embeddable Google Form capturing name, phone, property interest, budget range, and preferred contact time.
- Real-time write of form submissions to a dedicated Google Sheet (no shared spreadsheet across clients).
- Automated first-contact SMS within 60 seconds of form submission.
- A two-step reminder sequence (24h, 72h) for leads that have not replied or booked.
- Inbound SMS reply handling: Claude-drafted responses scoped to property Q&A and scheduling logistics, written to the Sheet as a pending draft.
- An owner approval queue (web dashboard) where the broker reviews and explicitly sends any AI-drafted reply.

- Hard escalation to a human number for financing/legal/negotiation questions or low-confidence replies.
- A dashboard showing lead status, last contact time, next scheduled action, and average first-response time.

3.2 Out of Scope (Phase 1)

- Auto-sending AI-drafted replies without broker approval.
- Multi-tenant infrastructure (Phase 1 targets one test client; scaling is a deliberate Phase 2 step).
- CRM integration, MLS/listing-data integration, or e-signature/offer workflows.
- WhatsApp Business API (documented as a drop-in alternative to Twilio SMS, not built in Phase 1).

4. Stakeholders

Role	Interest
Broker / Brokerage Owner	Primary user; approves AI drafts; owns the Twilio/Claude accounts and credentials.
Lead (prospective buyer/renter/seller)	Receives first contact, reminders, and (once approved) AI-assisted replies.
Solution Implementer	Deploys and configures the Apps Script project, Form, Sheet, and Twilio number per client.

5. Functional Requirements

5.1 Intake Layer

ID	Requirement
FR-1	The system shall present a Google Form with fields: name, phone, property interest, budget range, preferred contact time.
FR-2	The system shall write each submission to a dedicated Google Sheet in real time via the native Forms→Sheets integration.
FR-3	The system shall not collect a Google-verified email address as a condition of submitting the form.

5.2 Trigger & Orchestration Layer

ID	Requirement
FR-4	On new form submission, the system shall format and send a first-contact SMS within 60 seconds.
FR-5	The system shall log contact status (contacted / pending / booked / escalated / closed) back to the Sheet.
FR-6	The system shall run a recurring reminder sweep (default: every 15 minutes) that sends the next reminder in the sequence to any lead whose next-action time has passed and who has not replied or booked.
FR-7	The system shall stop the reminder sequence once a lead replies, books, or is escalated.

5.3 Phone/Messaging Layer

ID	Requirement
FR-8	The system shall send outbound SMS via Twilio using per-client credentials.
FR-9	The system shall route inbound SMS replies to a Claude API call scoped by a system prompt to property Q&A and scheduling logistics only.
FR-10	The system shall hard-escalate to a human phone number (via SMS alert) when the model is not confident, the message concerns financing/legal/negotiation, or the lead explicitly asks for a human.
FR-11	The system shall never send an AI-drafted reply to a lead automatically; every draft requires explicit broker approval.

5.4 Dashboard Layer

ID	Requirement
FR-12	The system shall provide a web dashboard showing, per lead: status, last contact time, next scheduled action, and response-time metric.
FR-13	The system shall let the broker approve a pending AI draft with one action, which sends the SMS and updates the Sheet.
FR-14	The dashboard shall be restricted to the broker's Google account (or domain), separate from the public Twilio webhook endpoint.

6. Non-Functional Requirements

Category	Requirement
Performance	First-contact SMS shall be sent within 60 seconds of form submission under normal Apps Script quota conditions.
Security — Isolation	Each client shall have a fully separate Spreadsheet and Apps Script project; no data is shared across clients in Phase 1.
Security — Secrets	Twilio and Claude credentials shall be stored in Apps Script Script Properties, never in the Sheet or client-side/dashboard code.
Security — PII	No PII shall appear in URL query parameters; the inbound Twilio webhook uses POST only. Logged phone numbers are masked to the last 4 digits.
Security — Inbound auth	The public webhook shall require a shared-secret token, since Apps Script cannot read the standard Twilio signature header.
Availability	Built on Google Apps Script and Twilio's published SLAs; no additional hosting is introduced for the MVP.
Auditability	Every inbound/outbound message shall be appended to a per-lead conversation log in the Sheet.

7. System Architecture Overview

The MVP is built on Google Apps Script bound to a Google Sheet, chosen because it provides, without standing up separate infrastructure: native Google Form→Sheet writes, cron-equivalent scheduling via time-driven triggers, a

public webhook endpoint for inbound Twilio messages, and a private web-app endpoint for the owner-facing approval dashboard.

- Google Form → Google Sheet (native write).
- Installable "on form submit" trigger → builds and sends first-contact SMS → logs status.
- Recurring time-driven trigger → scans Sheet for due reminders → sends and reschedules.
- Twilio inbound webhook (public deployment) → matches phone to lead row → Claude API drafts a reply → written to Sheet, unsent.
- Owner dashboard (private deployment, Google-account gated) → approval queue → Approve & Send calls Twilio directly.

8. Data Requirements (Leads Sheet Schema)

Column	Field
A–F	Timestamp, Name, Phone, Property Interest, Budget Range, Preferred Contact Time (from Form)
G	Status (pending / contacted / replied / reminder_sent / escalated / booked / closed)
H–I	LastContactAt, NextActionAt
J	ResponseTimeSeconds
K–L	AI DraftReply, ConversationLog
M–N	Approved (checkbox), ClientId

9. Assumptions & Constraints

- The broker holds (or will create) their own Google, Twilio, and Anthropic accounts; credentials are not shared with or stored by the implementer.
- US SMS is subject to A2P 10DLC registration and STOP/HELP opt-out handling at the Twilio/carrier level; the MVP does not implement its own opt-out list.
- Apps Script has a per-project trigger quota (20), which is why reminders use one recurring sweep rather than one trigger per lead.

10. Risks & Mitigations

Risk	Mitigation
AI drafts an inaccurate or overconfident reply.	All drafts require explicit broker approval before sending; low-confidence replies hard-escalate instead of guessing.
Public webhook abused to inject fake inbound messages.	Shared-secret token required on every inbound request; token is rotatable.
Apps Script quota limits at higher volume.	Documented ceiling in the setup guide; n8n/Make called out as a Phase 2 alternative if outgrown.

11. Success Metrics (MVP Acceptance Criteria)

- A test lead submitted via the Form receives a first-contact SMS in under 60 seconds.
- A reply from the test lead produces a drafted AI response visible in the dashboard, unsent.
- Approving the draft in the dashboard sends the SMS and updates the Sheet's status/approval fields.
- A message containing financing/negotiation content is escalated to the broker's phone instead of receiving an AI draft.
- The dashboard accurately reflects status, last contact, next action, and average response time.

12. Phase 2 (Out of Scope, Noted for Roadmap)

Multi-tenant scaling by duplicating the Apps Script project + Sheet per client; optional migration to n8n/Make if trigger-quota ceilings are reached; optional move from an approval-queue trust model to selective auto-send once reply quality is proven.